

Complex Number 21

1. Let w be an n^{th} root of unity. Find the value of $\sum_{k=0}^n \left(w^k + \frac{1}{w^k} \right)$.
2. Show that $(1 + \cos \theta + i \sin \theta)^n = 2^n \cos^n \left(\frac{\theta}{2} \right) \left[\cos \frac{n\theta}{2} + i \sin \frac{n\theta}{2} \right]$, where n is a positive integer.
3. Express the roots of the following equations in $x + iy$ form:
 - a. $z^2 - (1 - 4i)z - (5 - i) = 0$
 - b. $z^2 + (2 + 4i)z - 11 - 2i = 0$
 - c. $z^2 + (4 + 2i)z + (3 + 2i) = 0$
 - d. $z^2 + (4 - 2i)z + 6 = 0$
4. If $z = \cos \theta + i \sin \theta$, prove that $\frac{2}{1+z} = 1 - i \tan \frac{\theta}{2}$.
5. a. Show that $(1 + i)$ is a root of the polynomial $P(x) = x^3 + x^2 - 4x + 6$.
b. Hence resolve $P(x)$ into irreducible factors over the complex field.
6. Find the fourth roots of $2\sqrt{3} + 2i$.
7. A complex root of $z^3 + z^2 + z - 39 = 0$ is known to be $z = 3i - 2$. Deduce the value of the other two roots.
8. a. If $z = (\cos \theta + i \sin \theta)$, use De Moivre's Theorem to prove that $\left(z^n + \frac{1}{z^n} \right) = 2 \cos n\theta$.
b. Hence, or otherwise, prove that $8 \cos^4 \theta = \cos 4\theta + 4 \cos 2\theta + 3$.
9. The point A, B and C represent z_1, z_2 and $z_1 + z_2$ in an Argand diagram where z_1 and z_2 satisfy the relationship $\frac{z_2 - z_1}{z_2 + z_1} = ki$ where k is a positive number (and $i = \sqrt{-1}$).
 - a. Illustrate such an arrangement on an Argand diagram, denoting clearly properties of the quadrilateral $OACB$ and its diagonals (O is the origin).
 - b. What is the shape of the quadrilateral $OACB$ if
 - i. $k = 1$
 - ii. $k \neq 1$
 - c. Write down the area of the quadrilateral $OACB$ in terms of z_1 and z_2 .
10. An isosceles triangle ABP in the Argand diagram, with base AB and $\angle APB = \alpha$. PM is the perpendicular bisector of AB and so bisects $\angle APB$. Given that A and B represent the complex numbers z_1 and z_2 respectively, find, in terms of z_1, z_2 and α , the complex numbers represented by:
 - a. The vector AM .
 - b. The vector MP .Hence show that P represents the complex number $\frac{1}{2} \left(1 - i \cot \frac{\alpha}{2} \right) z_1 + \frac{1}{2} \left(1 + i \cot \frac{\alpha}{2} \right) z_2$.
11. a. Sketch nearly on the same diagram the loci of z when $|z - (3 + 2i)| = 2$ and $|z + 3| = |z - 5|$.
[clearly label each diagram]
b. Hence write down all the values of z which satisfy simultaneously: $|z - (3 + 2i)| = 2$ and $|z + 3| = |z - 5|$.
c. Determine the values of k for which the simultaneous equation: $|z - (3 + 2i)| = 2$ and $|z - 2i| = k$ have exactly one solution for z .
d. For the locus of z when $|z - (3 + 2i)| = 2$, find the maximum value of $\arg z$.
12. Given $z = \cos \theta + i \sin \theta$ and that m and n are integers.
 - a. Show that $\frac{z^m + z^{-m}}{2} = \cos m\theta$.
 - b. Hence, using part a., show that $\cos m\theta \times \cos n\theta = \frac{1}{2} [\cos(m + n)\theta + \cos(m - n)\theta]$.
13. Let $P(z) = z^8 - \frac{5}{2}z^4 + 1$. The complex number α is a root of $P(z) = 0$.

- a. Show that $i\alpha$ and $\frac{1}{\alpha}$ are also roots of $P(z) = 0$.
 b. Find one of the roots of $P(z) = 0$ in exact form.
 c. Hence solve fully $P(z) = 0$.

14. a. Show that

$$\cos \theta + \cos \phi = 2 \cos \left(\frac{\theta + \phi}{2} \right) \cos \left(\frac{\theta - \phi}{2} \right)$$

$$\sin \theta + \sin \phi = 2 \sin \left(\frac{\theta + \phi}{2} \right) \cos \left(\frac{\theta - \phi}{2} \right)$$

b. Hence show that $e^{i\theta} + e^{i\phi} = 2 \cos \left(\frac{\theta - \phi}{2} \right) e^{\frac{1}{2}i(\theta + \phi)}$

15. Show that the roots of $(z + 1)^8 - z^8 = 0$ are $z = -\frac{1}{2}, -\frac{1}{2} \left(1 \pm i \cot \left(\frac{k\pi}{8} \right) \right)$, where $k = 1, 2, 3$.

16. If k is a multiple of 4, prove that $(-1 + i)^k$ is real

17. a. Find the minimum value of the positive integer m for which $(\sqrt{3} + i)^m$ is: i. real, ii. purely imaginary.

b. Evaluate $(\sqrt{3} + i)^m$ for each of the above values of m

18. a. i. Use De Moivre's Theorem to find the formula for $\cos 6\theta$. (You may leave your answer in terms of mixtures of $\cos \theta$ and $\sin \theta$.)

ii. Hence show that $\cos 6\theta = (\cos^2 \theta - \sin^2 \theta)(\cos^4 \theta - 14 \cos^2 \theta \sin^2 \theta + \sin^4 \theta)$

iii. Hence using that $\cos \frac{\pi}{2} = 0$ show that $\cos \frac{\pi}{12}$ is a root of the equation $16x^4 - 16x^2 + 1 = 0$

iv. Hence find the exact value of $\cos \frac{\pi}{12}$

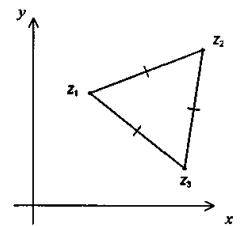
b. i. Resolve $z^6 + 1$ into real quadratic factors.

ii. Hence show that $\cos 3\theta = 4 \cos \theta \left(\cos^2 \theta - \cos^2 \frac{\pi}{6} \right)$

19. If complex number z_1, z_2, z_3 represent vertices of an equilateral triangle.

a. Express $z_2 - z_1$ in terms of $z_3 - z_1$

b. Hence prove $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$



20. Let w be an n^{th} root of unity. Show that $(1 + 2w + 3w^2 + 4w^3 + \dots + nw^{n-1})(w - 1) = n$.

21. Find all the roots of $(z + 1)^n + (z - 1)^n = 0$ and show that they are purely imaginary.

22. Let w be the $(n + 1)^{th}$ root of unity. Show that $\sum_{k=1}^n \frac{1}{1 - w^k} = \frac{n}{2}$.

23. Find the value of $\sum_{k=0}^n \cos \left(\frac{2k\pi}{2n+1} \right)$.

Answer

1. 2

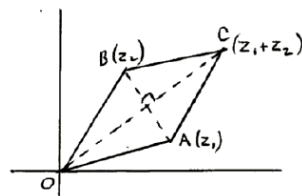
3. a. $2 - 3i, -1 - i$ b. $2 - i, -4 - 3i$ c. $-1, -3 - 2i$ d. $-1 - i, -3 + 3i$

5. b. $P(x) = (x + 3)\{x - (1 + i)\}\{x - (1 - i)\}$

6. $2\text{cis} \frac{\pi}{24}, 2\text{cis} \frac{13\pi}{24}, 2\text{cis} \frac{-11\pi}{24}, 2\text{cis} \frac{-23\pi}{24}$

7. $-2 - 3i, 3$

9. a. $OACB$ is a rhombus. b. i. $k = 1 \rightarrow \text{Square}$, ii. $k \neq 1 \rightarrow \text{Rhombus}$ c. Area = $\frac{1}{2}|z_1 + z_2||z_2 - z_1|$



10. a. AM represents $\frac{z_2 - z_1}{2}$ b. MP represents $\left(\frac{z_2 - z_1}{2} \right) i \cot \frac{\alpha}{2}$

11. a. $x = 1, (x - 3)^2 + (y - 2)^2 = 4$ b. $z = 1 + 2i$ c. $k = 1$ or $k = 5$ d. $\max \arg(z) = 2 \tan^{-1} \left(\frac{2}{3} \right)$

13. b. $z = \sqrt[4]{2}$ c. the roots are $\pm \sqrt[4]{2}, \pm \sqrt[4]{2}i, \pm \frac{1}{\sqrt[4]{2}}, \pm \frac{i}{\sqrt[4]{2}}$

14. prove

17. a. i. 6 ii. 3 b. $-64, 8i$

18. a. i. $\cos 6\theta = \cos^6 \theta - 15 \cos^4 \theta \sin^2 \theta + 15 \cos^2 \theta \sin^4 \theta - \sin^6 \theta$ iv. $\cos \frac{\pi}{12} = \frac{\sqrt{2 + \sqrt{3}}}{2}$ b. i. $z^6 + 1 = (z^2 - 2z \cos \frac{\pi}{6} + 1)(z^2 - 2z \cos \frac{5\pi}{6} + 1)$

19. a. $z_2 - z_1 = (z_3 - z_1) \text{cis} \frac{\pi}{3}$

20. prove

21. prove

22. prove

23. $\frac{1}{2}$